Chapter 13: What are the Chances?

The chance of something happening gives the percentage of time it is expected to happen, when the basic process is done over and over again, independently and under the same conditions.

Examples:

- Tossing a coin:
  Chance of getting heads is 1 in 2, or 50%.
- Rolling a die:
  Chance of rolling a 3 spot is 1 in 6, or $16\frac{2}{3}\%$.
- Shuffling a deck of cards and dealing out the top 5:
  Chance of getting a full house (one pair and three of a kind) is

$$\frac{3{,}744}{2{,}598{,}960} = 0.0014406 \approx 0.14\% \text{ (Why?)}$$

Poker Cards

Many examples involve cards.

- A deck of cards has 4 suits:

$$\begin{array}{cccc} \text{Clubs} & \text{Diamonds} & \text{Hearts} & \text{Spades} \\ \clubsuit & \diamondsuit & \heartsuit & \spadesuit \end{array}$$

- There are 13 cards in each suit:

$$\begin{array}{ccccccccccccc} \text{A,} & \text{2,} & \text{3,} & \text{4,} & \text{5,} & \text{6,} & \text{7,} & \text{8,} & \text{9,} & \text{10,} & \text{J,} & \text{Q,} & \text{K} \\ \downarrow & & & & & & & & & & \downarrow & \downarrow & \downarrow \\ \text{ace} & & & & & & & & & & \text{jack} & \text{queen} & \text{king} \end{array}$$

- So, there are $4 \times 13 = 52$ cards in the deck.

Chances are between 0% and 100%.

- Something that's impossible happens _____ of the time.
- Something that's sure to happen occurs _______ of the time.

Chances can be expressed in various ways. If in the long run some thing happens 3 times out of 4, it's chances are:

• 3 out of 4 • 3 in 4 • 3/4 • 0.75 • 75%.

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13.1 Complementation Rule

$$\boxed{\begin{array}{c}\text{The chance of something happening}\\ = 100\% - \text{the chance of the opposite thing.}\end{array}}$$

Example. In poker (five-card draw), the chance of being dealt a full house (one pair and three of a kind) is about 0.14%. The chance of NOT being dealt a full house is then

$$\begin{aligned} &100\% - \text{the chance of being dealt a full house} \\ = \ &100\% - 0.14\% = 99.86\%. \end{aligned}$$

$$\boxed{\begin{array}{c}\text{The chance of something happening} \\ = 100\% - \text{the chance of the opposite thing.}\end{array}}$$

- useful when the opposite thing has a simple description

  — so its chance is easy to compute,

  while the thing itself has a complex description

  — so its chance is hard to compute directly.

Example: what is the chance that in 4 tosses of a fair coin, there is at least one head?

- the opposite thing: no heads in 4 tosses
- Out of the 16 possible outcomes that could happen, just one has no heads (all tails). So the chance of the opposite thing is $\frac{1}{16} = 0.0625 = 6.25\%$.
- The chance of at least one head is $100\% - 6.25\% = 93.75\%$.

If a chance process is sure to turn out in some one of several **equally likely** ways, and if a thing of interest happens in some of those ways (the "favorable" outcomes) but fails in the other ways (the "unfavorable" outcomes), then the chance that the thing happens equals the ratio

$$\frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}.$$

Example: A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that the 2nd card is a king?

- There are ______ outcomes for the 2nd card, and they are all equally likely.
- _____ of those outcomes are kings.
- The chance that the 2nd card is a king equals ________ or _______%.

If a chance process is sure to turn out in some one of several **equally likely** ways, and if a thing of interest happens in some of those ways (the "favorable" outcomes) but fails in the other ways (the "unfavorable" outcomes), then the chance that the thing happens equals the ratio

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- There are __52__ outcomes for the 2nd card, and they are all equally likely.
- __4__ of those outcomes are kings.
- The chance that the 2nd card is a king equals __4/52__ or __7.7__%.

If a chance process is sure to turn out in some one of several **equally likely** ways, and if a thing of interest happens in some of those ways (the "favorable" possibilities) but fails in the other ways (the "unfavorable" possibilities), then the chance that the thing happens equals the ratio

$$\frac{\text{number of favorable possibilities}}{\text{total number of possibilities}}.$$

Caution

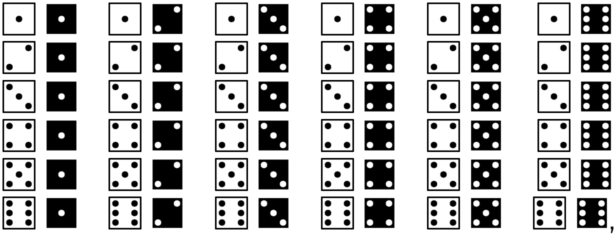
Before applying this rule, make sure that all those ways are **equally likely**.

See the example on the next page.

A white die and a black die are to be thrown. To find the chance of getting a total of 7 spots, which of the following arguments is correct?

Argument 1 There are 11 ways: 2,3,4,5,6,7,8,9,10,11, and 12, that the total spots of two dice can come out, and one of them is 7, so the chance is $1/11$.

Argument 2 There are 36 ways for the white and black dice to fall:



and 6 of them has a total of 7 spots:

so the chance is $6/36 = 1/6$.

13.3 Conditional Probabilities

A deck of cards is shuffled and the two top cards are placed face down on a table. The 1st card is turned over. It is a king. What is the chance that the 2nd card is a king?

- Given that the 1st card is a king:
  - There are now ______ outcomes for the 2nd card, all of which are equally likely.
  - _____ of those outcomes are kings.
- The **conditional chance** that the 2nd card is a king, **given** that the 1st card is a king, equals _____ in ______, or 5.9%.

What is the chance that the 2nd card is a king if the 1st card is *not* a king?

- Given that the 1st card isn't a king:
  - There are ______ equally likely outcomes for the 2nd card.
  - _____ of those outcomes are kings.
- The conditional chance that the 2nd card is a king, given that the 1st card is not, equals _____ in ______, or 7.8%

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  - There are now __51__ outcomes for the 2nd card, all of which are equally likely.
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  - There are __51__ equally likely outcomes for the 2nd card.
  - __4__ of those outcomes are kings.
- The conditional chance that the 2nd card is a king, given that the 1st card is not, equals __4__ in __51__, or 7.8%

13.4 The Multiplication Rule

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that both cards are kings?

- Imagine making many such deals.
  - The 1st card will be a king about ________ of the time.
  - Among the deals where the 1st card is king, the 2nd card will be a king about ________ of the time.
  - So both cards will be kings about ________ of ________ of the time.
- The chance that both cards are kings equals:

(The *unconditional* chance that the 1st card is a king)

$\times$ (the *conditional* chance that the 2nd card is a king *given* that the 1st card is a king)

$$= \frac{1}{13} \times \frac{3}{51} = \frac{1}{221} = 0.45\%.$$

13.4 The Multiplication Rule

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that both cards are kings?

- Imagine making many such deals.
  - The 1st card will be a king about __4/52__ of the time.
  - Among the deals where the 1st card is king, the 2nd card will be a king about ________ of the time.
  - So both cards will be kings about ________ of ________ of the time.
- The chance that both cards are kings equals:

(The *unconditional* chance that the 1st card is a king)

$\times$ (the *conditional* chance that the 2nd card is a king *given* that the 1st card is a king)

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- Imagine making many such deals.
  - The 1st card will be a king about $\underline{\quad 4/52 \quad}$ of the time.
  - Among the deals where the 1st card is king, the 2nd card will be a king about $\underline{\quad 3/51 \quad}$ of the time.
  - So both cards will be kings about ________ of ________ of the time.
- The chance that both cards are kings equals:

(The *unconditional* chance that the 1st card is a king)

× (the *conditional* chance that the 2nd card is a king *given* that the 1st card is a king)

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  - The 1st card will be a king about $\underline{\quad 4/52 \quad}$ of the time.
  - Among the deals where the 1st card is king, the 2nd card will be a king about $\underline{\quad 3/51 \quad}$ of the time.
  - So both cards will be kings about $\underline{\quad 3/51 \quad}$ of $\underline{\quad 4/52 \quad}$ of the time.
- The chance that both cards are kings equals:

(The *unconditional* chance that the 1st card is a king)

× (the *conditional* chance that the 2nd card is a king *given* that the 1st card is a king)

$$= \frac{1}{13} \times \frac{3}{51} = \frac{1}{221} = 0.45\%.$$

The Multiplication Rule (Cont'd)

$$\begin{aligned}&\text{(The chance that two things will both happen)}\\ =\ &\text{(the unconditional chance that the 1st will happen)}\\ &\quad\times\text{(the conditional chance that the 2nd will happen}\\ &\qquad\text{given that the 1st has happened).}\end{aligned}$$

In mathematical notation,

$$\boxed{\mathrm{P}(AB) = \mathrm{P}(A) \times \mathrm{P}(B|A)}$$

- $\mathrm{P}(A)$: the (unconditional) chance that A will happen
- $\mathrm{P}(A \text{ and } B)$ or $\mathrm{P}(AB)$: the chance that both A and B will happen ("and" is often omitted.)
- $\mathrm{P}(A|B)$: the conditional chance that A will happen given that B has happened

Here, "P" is short for "probability" (chance), and the vertical bar is read "given"

Example 1

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that neither card is a king?

- The chance that the 1st card is not a king = ________.
- Given that the 1st card is not a king, the conditional chance that the 2nd card is not a king = ________.
- So the chance that both cards are not kings = ________________, or 85.1%.

Example 1

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that neither card is a king?

- The chance that the 1st card is not a king = $\underline{\quad 48/52 \quad}$.
- Given that the 1st card is not a king, the conditional chance that the 2nd card is not a king = ________.
- So the chance that both cards are not kings = ________________, or 85.1%.

Example 1

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that neither card is a king?

- The chance that the 1st card is not a king = $\underline{\quad 48/52 \quad}$.
- Given that the 1st card is not a king, the conditional chance that the 2nd card is not a king = $\underline{\quad 47/51 \quad}$.
- So the chance that both cards are not kings = ________________, or 85.1%.

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A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that neither card is a king?

- The chance that the 1st card is not a king $=$ $\underline{\quad 48/52 \quad}$.
- Given that the 1st card is not a king, the conditional chance that the 2nd card is not a king $=$ $\underline{\quad 47/51 \quad}$.
- So the chance that both cards are not kings $=$ $\underline{\quad 48/52 \times 47/51 \quad}$, or 85.1%.

Example 2

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that at least one card is a king?

- The chance of the opposite thing — that neither card is a king — equals __________.
- By the complementation rule, the chance that at least one card is a king equals ___________________________.
- Note: This is not equal to the chance that 1st card is a king, plus the chance that the 2nd card is a king

$$7.7\% + 7.7\% = 15.4\% \neq 14.9\%$$

Example 2

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that at least one card is a king?

- The chance of the opposite thing — that neither card is a king — equals __85.1%__.
- By the complementation rule, the chance that at least one card is a king equals __________________.
- Note: This is not equal to the chance that 1st card is a king, plus the chance that the 2nd card is a king

$$7.7\% + 7.7\% = 15.4\% \neq 14.9\%$$

Example 2

A deck of cards is shuffled and the two top cards are placed face down on a table. What is the chance that at least one card is a king?

- The chance of the opposite thing — that neither card is a king — equals $\underline{\quad 85.1\% \quad}$.
- By the complementation rule, the chance that at least one card is a king equals $\underline{\quad 100\% - 85.1\% = 14.9\% \quad}$.
- Note: This is not equal to the chance that 1st card is a king, plus the chance that the 2nd card is a king

$$7.7\% + 7.7\% = 15.4\% \neq 14.9\%$$

Example 3

A die is rolled twice. What is the chance of getting two aces? (An ace is a 1-spot.)

- The chance of an ace on the 1st roll equals _______ .
- Given an ace on the 1st roll, the conditional chance of an ace on the 2nd roll equals _______ .
- So the chance of getting two aces equals
  _______ $\times$ _______ = ________.

Example 3

A die is rolled twice. What is the chance of getting two aces? (An ace is a 1-spot.)

- The chance of an ace on the 1st roll equals $\underline{\quad 1/6 \quad}$.
- Given an ace on the 1st roll, the conditional chance of an ace on the 2nd roll equals _______ .
- So the chance of getting two aces equals

  _______ $\times$ _______ = ________.

Example 3

A die is rolled twice. What is the chance of getting two aces? (An ace is a 1-spot.)

- The chance of an ace on the 1st roll equals $\underline{\quad 1/6 \quad}$.
- Given an ace on the 1st roll, the conditional chance of an ace on the 2nd roll equals $\underline{\quad 1/6 \quad}$.
- So the chance of getting two aces equals
  $\underline{\qquad\qquad} \times \underline{\qquad\qquad} = \underline{\qquad\qquad}$.

Example 3

A die is rolled twice. What is the chance of getting two aces? (An ace is a 1-spot.)

- The chance of an ace on the 1st roll equals $\underline{\quad 1/6 \quad}$.
- Given an ace on the 1st roll, the conditional chance of an ace on the 2nd roll equals $\underline{\quad 1/6 \quad}$.
- So the chance of getting two aces equals $\underline{\quad 1/6 \quad} \times \underline{\quad 1/6 \quad} = \underline{\quad 1/36 \quad}$.

Some Note on the Multiplication of Chances

When multiplying chances, it is safest to work with fractions or decimals, rather than percents.
For example, if the 1st chance is 1 in 2, and the 2nd is 3 in 5, then the product is

$$\frac{1}{2} \times \frac{3}{5} = \frac{1 \times 3}{2 \times 5} = \frac{3}{10}, \quad \text{or} \quad 0.5 \times 0.6 = 0.30.$$

The equivalent calculation with percents is

$$50\% \text{ of } 60\% = 30\% \quad (\text{or } 60\% \text{ of } 50\% = 30\%),$$

not

$$50\% \times 60\% = 3000\%^2$$

which is nonsense.

Multiplication Rule for Three Things

Is there a multiplication rule for finding the chance that three things will all happen?

- Yes. Start with the unconditional chance that the 1st thing will happen.
- Multiply that by the conditional chance that the 2nd thing will happen given that the 1st thing has happened.
- Multiply again, by the conditional chance that the 3rd thing will happen given that the first two things have happened.

In mathematical notation,

$$\mathrm{P}(ABC) = \mathrm{P}(A) \times \mathrm{P}(B|A) \times \mathrm{P}(C|AB)$$

Full House

In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

<u>Idea</u>: 5 things must happen:

Card	Restriction	Chance
1st	none	_____ in _____(unconditional)
2nd	same rank as the 1st	_____ in _____(conditional)
3rd	a different rank than the 1st	_____ in _____(conditional)
4th	same rank as the 3rd	_____ in _____(conditional)
5th	same rank as the 3rd	_____ in _____(conditional)

By the multiplication rule, the sought after chance equals

$$\frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48}, \text{ or } 0.000144, \text{ or } 0.0144\%.$$

Q: In poker, what's the chance of being dealt a full house (one pair and three of a kind)?

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In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

<u>Idea</u>: 5 things must happen:

Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	____ in ____ (conditional)
3rd	a different rank than the 1st	____ in ____ (conditional)
4th	same rank as the 3rd	____ in ____ (conditional)
5th	same rank as the 3rd	____ in ____ (conditional)

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<u>Idea</u>: 5 things must happen:

Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	3 in 51 (conditional)
3rd	a different rank than the 1st	____ in ____ (conditional)
4th	same rank as the 3rd	____ in ____ (conditional)
5th	same rank as the 3rd	____ in ____ (conditional)

By the multiplication rule, the sought after chance equals

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In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

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Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	3 in 51 (conditional)
3rd	a different rank than the 1st	48 in 50 (conditional)
4th	same rank as the 3rd	____ in ____ (conditional)
5th	same rank as the 3rd	____ in ____ (conditional)

By the multiplication rule, the sought after chance equals

$$\frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48}, \text{ or } 0.000144, \text{ or } 0.0144\%.$$

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Full House

In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

Idea: 5 things must happen:

Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	3 in 51 (conditional)
3rd	a different rank than the 1st	48 in 50 (conditional)
4th	same rank as the 3rd	3 in 49 (conditional)
5th	same rank as the 3rd	____ in ____ (conditional)

By the multiplication rule, the sought after chance equals

$$\frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48}, \text{ or } 0.000144, \text{ or } 0.0144\%.$$

Q: In poker, what's the chance of being dealt a full house (one pair and three of a kind)?

Full House

In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

Idea: 5 things must happen:

Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	3 in 51 (conditional)
3rd	a different rank than the 1st	48 in 50 (conditional)
4th	same rank as the 3rd	3 in 49 (conditional)
5th	same rank as the 3rd	2 in 48 (conditional)

By the multiplication rule, the sought after chance equals

$$\frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48}, \text{ or } 0.000144, \text{ or } 0.0144\%.$$

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Full House

In (five-card draw) poker, what's the chance of being dealt, in order, a pair followed by three of a kind?

Idea: 5 things must happen:

Card	Restriction	Chance
1st	none	52 in 52 (unconditional)
2nd	same rank as the 1st	3 in 51 (conditional)
3rd	a different rank than the 1st	48 in 50 (conditional)
4th	same rank as the 3rd	3 in 49 (conditional)
5th	same rank as the 3rd	2 in 48 (conditional)

By the multiplication rule, the sought after chance equals

$$\frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48}, \text{ or } 0.000144, \text{ or } 0.0144\%.$$

Q: In poker, what's the chance of being dealt a full house (one pair and three of a kind)? See Chapter 14.

13.5 Independence

Two things are **independent** if the chances for the 2nd one given the 1st are the same, no matter how the 1st one turns out. Otherwise, they are **dependent**.

In mathematical notation,

$$A \text{ and } B \text{ are independent if } \mathrm{P}(B|A) = \mathrm{P}(B)$$

An Example of Dependent Events

A deck of cards is shuffled and the two top cards are placed face down on a table.

- Event 1: the 1st card is a king.
- Event 2: the 2nd card is a king.

Q: Are these two events independent, or dependent?
A:

- Given that the 1st card is a king, the chance that the 2nd card is a king equals _____ / 51.
- Given that the 1st card is not a king, the chance that the 2nd card is a king equals _____ / 51.

The chances for the 2nd event change, depending on how the 1st event turns out. So the two events are ____________.

An Example of Dependent Events

A deck of cards is shuffled and the two top cards are placed face down on a table.

- Event 1: the 1st card is a king.
- Event 2: the 2nd card is a king.

Q: Are these two events independent, or dependent?
A:

- Given that the 1st card is a king, the chance that the 2nd card is a king equals __3__ / 51.
- Given that the 1st card is not a king, the chance that the 2nd card is a king equals _____ / 51.

The chances for the 2nd event change, depending on how the 1st event turns out. So the two events are ____________.

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- Given that the 1st card is a king, the chance that the 2nd card is a king equals __3__ / 51.
- Given that the 1st card is not a king, the chance that the 2nd card is a king equals __4__ / 51.

The chances for the 2nd event change, depending on how the 1st event turns out. So the two events are __________.

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The chances for the 2nd event change, depending on how the 1st event turns out. So the two events are __dependent__.

More Examples

- Someone is going to roll a die twice. Are the two rolls independent, or dependent?
  - No matter how the 1st roll turns out, the 2nd roll will give 1, 2, 3, 4, 5, or 6, with equal chances. So the two rolls are ______________ .
- When drawing at random from a box
  - with replacement, the draws are independent;
  - without replacement, the draws are dependent.

More Examples

- Someone is going to roll a die twice. Are the two rolls independent, or dependent?
  - No matter how the 1st roll turns out, the 2nd roll will give 1, 2, 3, 4, 5, or 6, with equal chances. So the two rolls are ___independent___ .
- When drawing at random from a box
  - with replacement, the draws are independent;
  - without replacement, the draws are dependent.

13.6 Multiplication Rule for Independent Events

If several events are independent,
the chance that all of them will happen
equals the product of their unconditional chances.

In mathematical notation, if A and B are independent, then

$$\mathrm{P}(AB) = \mathrm{P}(A) \times \mathrm{P}(B)$$

Example 1

Every day you buy a lottery ticket that offers 1 chance in 1000 of winning. What is the chance that you never win in 1000 plays?

The question asks for the chance of losing on each play.

- The plays are independent.
- Your chance of losing on any particular play equals ______ in _______ , or _______.
- Your chance of losing on all 1000 plays equals ____________, or 0.368 .

The chance that you win at least once in 1000 plays equals ____________, or 0.632 .

- This would be hard to compute directly.
- The complementation rule is useful here.

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- Your chance of losing on all 1000 plays equals ___$(0.999)^{1000}$___, or 0.368 .

The chance that you win at least once in 1000 plays equals ___$1 - 0.368$___, or 0.632 .

- This would be hard to compute directly.
- The complementation rule is useful here.

Example 2

A computer is about to select a person at random from the U.S. population. Consider two events

- The person ages 85+.
- The person is a male.

Q: Are these two events independent, or dependent?

- P(the person is male)$=$ the percentage of males in the U.S. $\approx$ 49.5%
- P(the person is male $|$ the person is 85+) $=$ the percentage of males among those age over 85 $\approx$ 36.5% — women tend to live longer
- So age and sex are ________________ in this context.

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Example 2

A computer is about to select a person at random from the U.S. population. Consider two events

- The person ages $85+$.
- The person is a male.

Q: Are these two events independent, or dependent?

- P(the person is male)$=$ the percentage of males in the U.S. $\approx$ 49.5%
- P(the person is male $|$ the person is $85+$) $=$ the percentage of males among those age over 85 $\approx$ 36.5% — women tend to live longer
- So age and sex are __dependent__ in this context.

Example 2 (Cont'd)

As estimated in 2020, of the U.S. population,

- 2.0% were 85 or older, and
- 49.5% were male.

True or False and explain: $0.495 \times 0.02 \approx 1\%$ of the U.S. population are males and age 85+.

Example 2 (Cont'd)

As estimated in 2020, of the U.S. population,

- 2.0% were 85 or older, and
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True or False and explain: $0.495 \times 0.02 \approx 1\%$ of the U.S. population are males and age 85+.

__False__. Among those age 85 or older, only 36.5% are male, not 49.5%. Of the U.S. population in 2020, only

$$P(M \textit{ and } 85+) = P(85+)P(M|85+) = 0.02 \times 0.365 \approx 0.73\%$$

were males age 85+.

Blindly multiplying unconditional probabilities can give the wrong answer.

Summary

- The **frequency theory** of chance applies most directly to chance processes which can be repeated over and over again, independently and under the same conditions.
- The chance of something says about what percentage of the time it is expected to happen, when the basic process is repeated over and over again.
- Chances are between 0% and 100%. Impossibility is represented by 0%, certainty by 100%.
- If a chance process is sure to turn out in some one of several equally likely ways, and if a thing of interest happens in some of those ways (the "favorable" possibilities) but fails in the other ways (the "unfavorable" possibilities), then the chance that the thing happens equals the ratio

$$\frac{\text{number of favorable possibilities}}{\text{total number of possibilities}}.$$

Summary (Cont'd)

- **Complementation Rule**: the chance of something equals 100% minus the chance of its opposite.

$$\mathrm{P}(A) = 100\% - \mathrm{P}(\text{not } A)$$

- **Multiplication Rule**: The chance that two things will both happen equals the (**unconditional**) chance that the 1st one will happen, multiplied by the **conditional** chance that the 2nd will happen given that the 1st has happened.

$$\mathrm{P}(AB) = \mathrm{P}(A) \times \mathrm{P}(B|A)$$

- Two things are **independent** if the chances for the 2nd one stay the same no matter how the 1st turns out.

$$\mathrm{P}(B|A) = \mathrm{P}(B)$$

- **Independence Rule**: If two things are independent, the chance that both will happen equals the product of their unconditional chances.